

Evaluating the Resistance of RAG Methods to Poisoning

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Abstract

Retrieval Augmented Generation (RAG) systems are vulnerable to corpus poisoning, in which misinformation in the retrieval corpus leads the system to produce incorrect answers. While prior work has demonstrated that such attacks are effective, less attention has been given to whether different retrieval architectures vary in their susceptibility. In this work, we compare five RAG retrieval strategies under controlled poisoning: sparse BM25, dense, hybrid, reranked, and query decomposition. Using a Natural Questions-based corpus of 3,000 passages with 15% synthetically poisoned content, we hold the generator and evaluation queries constant while varying only the retrieval method. We report results using two complementary metrics, Poisoned Passage Rate (PPR) and Answer Deviation Ratio (ADR). Sparse and hybrid retrieval show the strongest resistance to poisoning, while dense and especially reranked retrieval are significantly more vulnerable. These findings reveal a tradeoff between semantic similarity and adversarial resistance.

1 Problem Introduction

1.1 Background

Retrieval Augmented Generation (RAG) is a prevalent technique in natural language processing that improves LLMs by integrating retrieval-based methods to enhance the accuracy of large language models in tasks such as question answering, information retrieval, and text generation. Although, this hinders the generation of false information or hallucinations, it does not dismiss all types of concerns. For example, the reliance on external corpora has led to a new vulnerability, corpus poisoning. When the underlying knowledge base contains falsehoods, RAG systems are susceptible to generating outputs with incorrect claims, contradictory information, and poisoned evidence.

1.2 Motivation

Corpus poisoning as an attack can cause significant issues in real-world situations. An example of one of these could be a type of medical system using RAG over a medical knowledge base: if even a small fraction of the documents contains misinformation or incorrect claims, the system could confidently identify the wrong answer as a cited and correct response, potentially leading to a negative outcome such as a misdiagnosis. These mistakes can also be harder to identify than normal LLM hallucinations, since the poisoning-induced errors are grounded in the retrieved text and appear to be more credible.

1.3 Research Gap

While prior work has largely focused on hallucinations caused by insufficient or missing knowledge, less attention has been paid to how different RAG architectures respond when the retrieved information itself is compromised. This is an important question to answer, because there are few variables that can be altered when designing a RAG system, and the retrieval strategy is one of the most consequential of those. For a system designer tasked with choosing the RAG architecture or deciding whether to add a reranking stage, there is somewhat of a lack of resources in the current literature to review. If different retrieval architectures exhibit significantly different levels of susceptibility to poisoning, that can directly help designers make these decisions, especially for low trust environments where poisoning is a threat.

1.4 Project Overview

This project investigates how different RAG architectures behave under these conditions by injecting controlled false information into an otherwise clean corpus. By comparing how various RAG methods retrieve and use this poisoned data, we

aim to understand which designs are more effective at avoiding incorrect evidence and which are more likely to repeat false claims that appear credible. In order to directly measure the effect of the retrieval strategy, we hold the generator model, corpus, and evaluation queries constant across all experiments, while varying the retrieval strategy. We evaluate five retrieval methods: sparse BM25, dense, hybrid, reranked, and query decomposition, on both a clean and a synthetically poisoned version of the same corpus.

2 Related Work

Recent studies have highlighted security and robustness challenges in RAG systems.

2.1 PoisonedRAG

Algorithm 1 PoisonedRAG Black-Box Attack

Require: Questions $\{Q_i\}_{i=1}^M$, answers $\{R_i\}_{i=1}^M$
Require: Hyperparameters N, L, V , LLM $\mathcal{M}$
Ensure: Malicious text set $\mathcal{T}$

- 1: **for** $i = 1$ to M **do**
- 2: **for** $j = 1$ to N **do**
- 3: $I_i^j \leftarrow \text{TEXTGENERATION}(Q_i, R_i, \mathcal{M}, L, V)$
- 4: **end for**
- 5: **end for**
- 6: **return** $\{Q_i \oplus I_i^j\}$

(Zou et al., 2024) introduced PosionedRAG, a framework for knowledge poisoning attacks in retrieval-augmented generation. Their work demonstrated that injecting a small number of deliberately crafted poisoned texts into a knowledge corpus can mislead RAG models to produce incorrect, biased, or target-specific answers. The study formalized poisoning as an optimization problem and proposed heuristic strategies to satisfy both retrieval and effectiveness conditions, showing attack success rates of up to 90% under both black-box and white-box settings. Overall, this work showed that standard defenses are insufficient to prevent such attacks, highlighting a vulnerability that arises from reliance on external knowledge sources. This is closely related to our experiment, since it is one of the first and most popular papers to introduce corpus poisoning as a real and effective attack vector against RAG systems. PoisonedRAG crafts documents that are designed to be retrieved for a target query, while our work uses a simpler generation based poisoning strategy in order to measure the robustness of the different RAG architectures to that poisoning.

2.2 POISONCRAFT

Algorithm 2 POISONCRAFT Procedure

Require: Knowledge base size n , poisoning rate p
Require: Shadow query set S , topics J
Require: Top-frequency words F , top- E variants
Require: Suffix lengths $\{l_1, l_2, \dots, l_L\}$, batch size b
Ensure: Poisoned document set $P = \bigcup_{j=1}^J P_j$

- 1: Partition S into topic subsets $\{S_1, \dots, S_J\}$
- 2: **for** $j = 1$ to J **do**
- 3: $B_j \leftarrow n \cdot p \cdot \frac{|S_j|}{|S|}$
- 4: **end for**
- 5: Extract top- F frequent words from S to form C_{freq}
- 6: **for** $j = 1$ to J **do**
- 7: $P_j \leftarrow \emptyset$
- 8: **while** $|P_j| < B_j$ **do**
- 9: Sample batch $S_j^a \subseteq S_j$ of size b
- 10: **for each** $l_{\text{init}} \in \{l_1, l_2, \dots, l_L\}$ **do**
- 11: $C_{\text{init}} \leftarrow l_{\text{init}} \times !$
- 12: $C \leftarrow C_{\text{inject}} + C_{\text{freq}} + C_{\text{init}}$
- 13: $\{C_{\text{adv}}^1, \dots, C_{\text{adv}}^E\} \leftarrow \text{OPTIMIZE}(C, S_j^a)$
- 14: Add poisoned documents to P_j until reaching B_j
- 15: **end for**
- 16: **end while**
- 17: **end for**
- 18: **return** $P = \bigcup_{j=1}^J P_j$

POISONCRAFT (Shao et al., 2025) also explored these weaknesses in RAG systems by evaluating the impact of injection attacks on the retrieval corpus. Unlike prior attacks that require knowledge of the user’s query or depend on specific triggers, POISONCRAFT operates without access to the target query or its topic, deeming it a query-agnostic model that closely models real-world conditions. The authors showed that strategically crafted poisoned documents were consistently retrieved and incorporated into model responses, leading the system to generate incorrect or manipulated outputs despite the absence of query-specific optimization. Overall, this work further demonstrated an inherent vulnerability introduced by reliance on external corpora. POISONCRAFT’s generation strategy is similar to ours, generating poisoned passages without knowledge of specific evaluation queries, as opposed to the PoisonedRAG experiment. Poisoned documents remain focused on the same topics as the corpus, but not designed to answer any specific question.

3 Dataset

3.1 Overview

In this project, we evaluate the robustness of different RAG methods to corpus poisoning using a large-scale question-answering dataset, along with

a controlled number of synthetically corrupted documents. The primary dataset we use is the Natural Questions dataset (Kwiatkowski et al., 2019), which comprises hundreds of thousands of real user queries and Wikipedia snapshots. These snapshots contain fixed length passages with factual and declarative information, which makes it an ideal knowledge source for the RAG architectures we plan to evaluate. This corpus has been well established in literature as a QA benchmark as well as a clean corpus in a RAG pipeline (Cheng et al., 2025). For our experiments, we extract 3,000 unique passages from the dataset and set aside 500 QA pairs to use as evaluation queries. Each query used for testing has its own ground truth short answer that can be derived from the dataset’s passages.

3.2 Poisoned Corpus

To measure the impact of misinformation on our RAG pipeline, we introduce synthetically poisoned documents into the corpus that contain intentionally false or misleading information. These are designed in a way such that they still resemble legitimate documents but contain incorrect factual claims. This way, the poisoning attacks resemble a real-world scenario where misinformation may be present. We ensure to keep our poisoning strategy consistent, using targeted insertions into the corpus.

We inject poisoned documents into our corpus at a fixed ratio of 15% of to the total corpus size. This ratio is somewhat higher than those used in existing papers such as PoisonedRAG in order to simulate an environment that is likely to be untrustworthy and potentially has a significant fraction of incorrect or outdated documents.

4 Methodology

Given a natural language query, the goal of a RAG system is to retrieve the answer from a set of documents (the corpus). When the corpus is partially corrupted, the system may ingest that incorrect information from the corpus and include it in its answer to the query. Our goal is to measure and compare how susceptible different RAG retrieval techniques are when the corpus becomes poisoned. To accomplish this, we first construct a standard RAG pipeline with three stages: retrieval, context construction, and generation.

To analyze the effect of retrieval design, we use one language model, Ollama, for generation across

all experiments. This ensures that differences in the system behavior are due to the retrieval strategy. We evaluate multiple RAG retrieval methods: **sparse, dense, hybrid, re-ranked**, and **query decomposition**, all operating with the same corpus.

4.1 Retrieval and Generation Pipeline

For each query, the retrieval component selects the top- k passages from the indexed corpus according to the specified retrieval method, where k is a controlled hyperparameter. These passages are then concatenated in order to provide a context to the generator. The generator then produces an answer based on the query and the retrieved context. Using this pipeline, we compare how each RAG architecture performs when exposed to corrupted data versus the original clean corpus.

4.2 Poisoning Procedure

To evaluate robustness under corpus corruption, we introduce controlled factual poisoning into the retrieval corpus. Starting from clean passages in the Natural Questions dataset, we generate poisoned variants using a large language model. Each poisoned passage is created by rewriting the original passage to preserve fluency and topical relevance, while intentionally altering key factual details such as entities, dates, or relationships to produce an incorrect answer to the associated question.

We construct a new poisoned corpus using 3,000 clean passages and inject poisoned documents at a fixed ratio of 15%. For each selected passage, a corresponding poisoned version is generated and inserted into the corpus, resulting in both clean and corrupted variants of similar content. These poisoned passages are designed to remain semantically similar to the original text, increasing their likelihood of retrieval under embedding-based methods.

The poisoned and clean passages are combined and randomly shuffled to form the final corpus. Because poisoned passages are aligned with the original queries and indexed using the same embedding representation, they present a realistic and challenging adversarial setting in which misleading information can be retrieved alongside correct evidence.

4.3 Retrieval Methods

We evaluate five retrieval strategies:

- **Sparse Retrieval:** Implemented using BM25, which ranks documents based on lexical overlap between the query and passages.

- **Dense Retrieval:** Uses vector embeddings to retrieve passages based on semantic similarity between queries and documents.
- **Hybrid Retrieval:** Combines sparse and dense scores using a weighted fusion strategy to balance lexical and semantic signals.
- **Reranked Retrieval:** Applies a cross-encoder reranker to reorder the top candidates retrieved by a dense retriever.
- **Query Decomposition:** Splits complex queries into sub-queries, retrieves passages for each sub-query using dense retrieval, and aggregates the results.

Sparse retrieval is implemented using the `rank_bm25` library. Dense retrieval uses the `all-MiniLM-L6-v2` model as a sentence transformer with passages indexed using FAISS. Hybrid retrieval combines normalized BM25 and dense scores using a weighted fusion function. The reranked retrieval uses `/ms-marco-MiniLM-L-6-v2` as the cross-encoder, and applies over the top 20 candidates before selecting the final top-k. Query decomposition prompts the generator to split each query into two to three sub-questions before retrieving results for each.

4.4 Evaluation

To assess the robustness of these RAG architectures under corpus poisoning, we use a small set of predefined metrics:

1. **Poisoned Passage Rate (PPR):** This metric measures the proportion of passages that were retrieved in our pipeline that contain poisoned content among the top k results.
2. **Exact Match (EM) and Token-Level F1:** These are standard metrics for QA models, and identify the correctness of generated answers relative to the ground truth. (Gan et al., 2025)
3. **Answer Deviation Ratio (ADR):** This metric measures the proportion of queries for which the model produces a correct answer on the clean corpus but an incorrect answer using the poisoned corpus. This aids in quantifying the impact of the corpus poisoning.

By using both PPR and ADR, we are able to capture different levels of vulnerability, with the PPR measuring whether poisoned passages are being retrieved, and ADR measuring whether those passages actually corrupt the answer. This is important, because a high PPR does not necessarily correspond with a high ADR, due to the fact that a generator can override poisoned context using its existing knowledge. Furthermore, multiple clean passages in the top-k may lower the influence of a single poisoned one. Reporting both metrics allows us to distinguish the impact at retrieval-level and generation-level.

Together, these metrics capture both answer correctness and resistance to corrupted evidence, allowing us to evaluate not only performance degradation but also susceptibility to misleading retrieval.

4.5 Comparative Analysis

Each RAG architecture is evaluated both on the clean and poisoned corpus using the same set of queries. We compare the performance across the different architectures by measuring differences across the three defined metrics. This comparison allows us to identify which RAG strategies are more resilient to corpus poisoning, and which ones are more likely to include incorrect evidence.

5 Results

Table 1 summarizes the performance of each retrieval method on both clean and poisoned corpora, using the described metrics.

Method	Corpus	EM	ROUGE-L	PPR	ADR _R
Sparse	clean	0.1300	0.2532	0.0000	—
	poisoned	0.1133	0.2305	0.1080	0.1702
Dense	clean	0.1167	0.2405	0.0000	—
	poisoned	0.0933	0.2063	0.1413	0.2527
Hybrid	clean	0.1267	0.2379	0.0000	—
	poisoned	0.1133	0.2261	0.1207	0.1882
Reranked	clean	0.1367	0.2505	0.0000	—
	poisoned	0.1067	0.2143	0.1533	0.2955
Decomposed	clean	0.1200	0.2414	0.0000	—
	poisoned	0.0967	0.2076	0.1413	0.2273

Table 1: Performance of each retrieval method on clean and poisoned corpora.

5.1 Overall Performance Degradation

Across all methods, poisoning consistently degrades performance, reducing both Exact Match (EM) and ROUGE-L scores. The EM and ROUGE-L scores are relatively low across all methods on the clean corpus before incurring any poisoning.

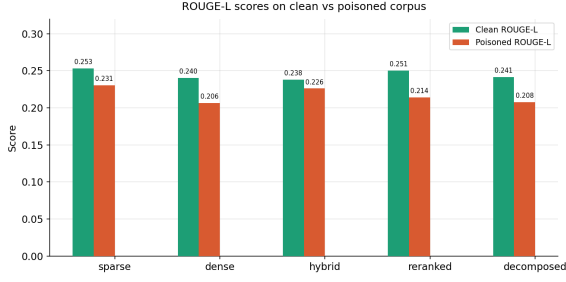


Figure 1: Rouge-L scores on clean versus poisoned corpus.

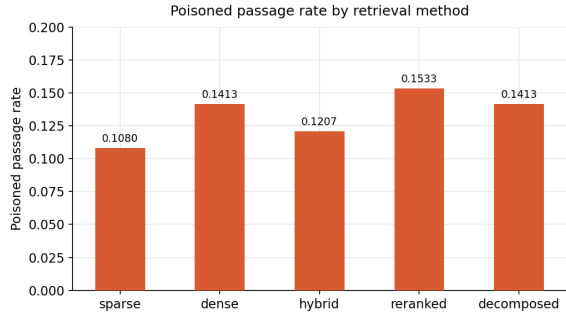


Figure 2: Poisoned passage rate by retrieval method.

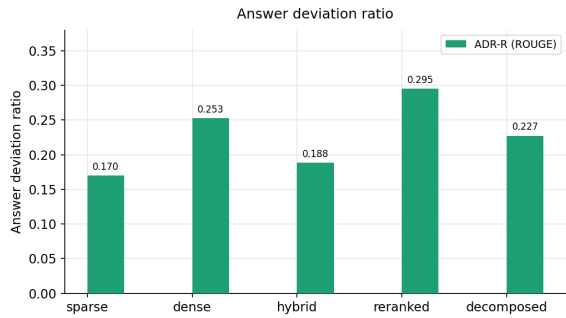


Figure 3: Answer deviation ratio by retrieval method.

This is consistent with known limitations of these metrics on the Natural Questions dataset, where ground truth answers are typically only one to five words. Exact match is especially harsh with shorter answers, and a correctly answered question with minor paraphrasing could score zero. ROUGE-L also generally struggles with shorter passages of text, since just a single word difference could disproportionately impact the score. Because of this, for this experiment we focus primarily on the degradation metrics, Δ ROUGE and ADR, measuring the impact of the poisoning rather than the quality of the model’s answers.

5.2 Retrieval Method Performance

Among the evaluated methods, **sparse** and **hybrid retrieval** exhibit the strongest resistance to RAG poisoning. Hybrid achieves the lowest ROUGE degradation (Δ ROUGE = 0.0118), while sparse maintains the lowest poisoned passage rate (PPR = 0.108) and relatively low answer deviation ratio (ADR = 0.170).

In contrast, **dense** and **reranked retrieval** are significantly more vulnerable. Reranked retrieval shows the highest degradation (Δ ROUGE = 0.0361) and the highest ADR (0.296), indicating that it is most likely to produce incorrect answers when exposed to poisoned data. Dense retrieval also performs poorly, with high PPR (0.141) and ADR (0.253).

Query decomposition demonstrates moderate resistance to poisoning. While it improves coverage of relevant information, it also increases exposure to poisoned passages, resulting in mid-range performance across all metrics (ADR = 0.227).

5.3 Metric Interpretation

Figures 2 and 3 also reveal a strong positive relationship between poisoned passage rate (PPR) and answer deviation ratio (ADR). Retrieval methods that expose the generator to more poisoned passages consistently produce larger deviations in final answers. For example, reranked retrieval exhibits both the highest PPR (0.1533) and the highest ADR (0.295), while sparse retrieval achieves the lowest values on both metrics.

Overall, the results demonstrate that corpus poisoning negatively impacts all evaluated RAG methods, but the magnitude of this degradation depends heavily on the retrieval architecture and its susceptibility to retrieving poisoned evidence.

6 Results Analysis

The observed differences in resistance to RAG poisoning can be attributed to how each retrieval method interacts with poisoned content.

Sparse retrieval (BM25) relies on lexical matching, which makes it less sensitive to semantically similar but factually incorrect passages. Since poisoned passages often modify factual details while preserving semantics, they are less likely to rank highly under sparse retrieval. Also, because BM25 scoring is based on term frequency and inverse document frequency, a poisoned passage that substitutes a correct entity name for an incorrect one could just lose the matching term entirely, sharply dropping its rank. This helps explain why sparse achieves the lowest PPR of 0.1080, significantly below the baseline of 0.15.

Dense retrieval, in contrast, relies on semantic similarity. Because poisoned passages are intentionally constructed to remain topically relevant, they often achieve high embedding similarity with the query. Also, embedding models like the one we used are trained to encode semantic meaning rather than factual correctness, so two passages that discuss the same topic but give different answers will be represented as very similar vectors. This means dense retrieval has no real way to distinguish a correct passage from a well-written incorrect one about the same subject, which leads to it being more likely to surface poisoned content, leading to higher PPR and ADR.

Hybrid retrieval balances these two approaches by combining lexical and semantic signals. This mitigates the weaknesses of dense retrieval while retaining its strengths, resulting in close to the best overall resistance among the evaluated methods. When a poisoned passage scores highly on the dense component, the Sparse BM25 component acts as a check on that result, and if the passage has altered key factual terms, its lexical score will become lower. This combination makes it harder for any single poisoned passage to dominate the top-k results, which helps explain why hybrid scores low.

Reranked retrieval amplifies the vulnerability of dense retrieval. This finding is somewhat counterintuitive, since reranking is understood to be an added quality improvement step. This is likely because the cross-encoder used in reranking assigns high scores to passages that appear semantically aligned with the query, even if they contain incorrect information. So the additional reranking

actually makes the retrieval even more susceptible to poisoned passages, explaining its worst performance out of the 5 strategies.

Query decomposition exhibits mixed behavior. Breaking queries into sub-questions, it improves retrieval coverage but also increases the number of retrieval operations. This expands the attack surface, making it more likely that at least one poisoned passage is retrieved. Additionally, since it relies on dense retrieval, it inherits its susceptibility to semantic poisoning.

These findings highlight a key tradeoff: methods that prioritize semantic understanding tend to be more vulnerable to poisoning, while those that rely on lexical matching are more resistant.

7 Limitations and Challenges

While our experiments produce a clear ordering of retrieval architectures by their robustness to corpus poisoning, several limitations affect the scope and interpretation of these findings. We discuss the most significant of these below to contextualize our results and motivate future work.

7.1 Poisoned Passage Generation

Generating passages that are both fluent, topically aligned, and factually incorrect is a non-trivial task. Our prompt-based strategy using GPT-OSS-120B produces natural, on-topic text, but the resulting corruptions vary in quality: some alter superficial details such as a single date or name, while others consist of multiple details being changed inconsistently. We did not perform a fine-grained quality audit, so our reported attack effectiveness reflects average behavior over a heterogeneous mix of corruption strengths. Our approach is also query-agnostic, in the sense of POISONCRAFT (Shao et al., 2025); stronger, query-aware attacks such as those used in PoisonedRAG (Zou et al., 2024) may produce different relative behaviors across architectures, particularly eroding sparse retrieval’s current advantage.

7.2 Evaluation Metric Limitations

Quantifying answer correctness against short ground-truth spans is harder than it appears. Exact Match is brittle to surface variation, and ROUGE-L is sensitive to the shortness of Natural Questions answers, where many spans are only a few tokens long and a single mismatch produces a disproportionate drop in score. Our Answer Deviation Ratio

measures relative change between clean and poisoned conditions on a per-query basis, but it inherits its underlying judgments from EM and ROUGE-L. A more semantically aware evaluation, such as LLM-as-a-judge scoring, would give us a more faithful picture of the actual impact of poisoning.

7.3 Generator Model Scale

We use Llama 3.2 3B as the sole generator across all experiments to isolate the effect of retrieval strategy. Smaller models tend to rely more heavily on retrieved context because they have less parametric knowledge to override misleading evidence, so our reported ADR values likely represent an upper bound on poisoning impact. We expect the relative ordering of retrieval methods to remain consistent across generator scales, but the absolute degradation in answer quality may shrink for stronger models. We leave verification with a larger generator to future work.

7.4 Scope of Experimental Setup

Our experiments are conducted on a single corpus, in English, with a fixed poisoning ratio of 15 percent. Wikipedia passages are unusually clean and stylistically uniform, which may make poisoned passages easier or harder to retrieve than they would be in a noisier real-world corpus, such as web-scraped news or user-generated content. The 15 percent ratio is intentionally elevated to simulate a low-trust environment, but the relative robustness of retrieval methods may shift at lower or higher levels of corruption; sparse retrieval’s advantage in particular may shrink as the volume of poisoned passages grows. We also evaluate only a single variant of each retrieval architecture, so our conclusions are best understood as broad qualitative trends rather than precise quantitative claims.

8 Conclusion

8.1 Summary

In this work, we investigated the robustness of different retrieval strategies in Retrieval-Augmented Generation (RAG) systems under corpus poisoning. By injecting synthetically corrupted passages into a Natural Questions-based corpus, we evaluated how five retrieval methods respond to misleading yet realistic information.

8.2 Key Findings

Our results show that retrieval design plays a critical role in system robustness. **Sparse** and **hybrid retrieval** methods demonstrate the strongest resistance to poisoning, exhibiting lower performance degradation and reduced exposure to poisoned passages. In contrast, **dense** and especially **reranked retrieval** are significantly more vulnerable, as semantic similarity allows poisoned passages to be retrieved and trusted more easily. **Query decomposition** provides moderate robustness, improving coverage but increasing exposure to corrupted content.

8.3 Tradeoffs

These findings highlight a key tradeoff: methods that emphasize semantic similarity improve retrieval quality under clean conditions but are more susceptible to adversarial manipulation, while lexical-based approaches offer greater robustness.

8.4 Implications and Future Work

This work highlights the importance of retrieval design in building reliable RAG systems and suggests that hybrid approaches are a promising direction for balancing performance and resistance. Future work could explore adaptive defenses, poisoning detection mechanisms, and robustness under varying levels and types of adversarial corruption.

References

- Mingyue Cheng, Yucong Luo, Jie Ouyang, Qi Liu, Huijie Liu, Li Li, Shuo Yu, Bohou Zhang, Jiawei Cao, Jie Ma, Daoyu Wang, and Enhong Chen. 2025. [A survey on knowledge-oriented retrieval-augmented generation](#).
- Aoran Gan, Hao Yu, Kai Zhang, Wenyu Yan, Zhenya Huang, Shiwei Tong, Enhong Chen, and Guoping Hu. 2025. [Retrieval augmented generation evaluation in the era of large language models: A comprehensive survey](#).
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. [Natural questions: A benchmark for question answering research](#). *Transactions of the Association for Computational Linguistics*, 7:452–466.
- Yangguang Shao, Xinjie Lin, Haozheng Luo, Chengshang Hou, Gang Xiong, Jiahao Yu, and Junzheng Shi. 2025. [Poisoncraft: Practical poisoning of](#)

retrieval-augmented generation for large language models.

Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. 2024. [Poisonedrag: Knowledge corruption attacks to retrieval-augmented generation of large language models](#).